PsyClick: Clinical Decision Support System for Detecting Single-Session Psychomotor Disturbances via EWMA-Based Multivariate Analysis of Keystroke and Cursor Dynamics

by

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Technical Documentation

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1. Purpose and Scope

This document is the consolidated Technical Documentation for PsyClick Secure, required under the CS0029 Final Project Specification (Section IV.B, item 3). It contains the System Architecture Diagram, Database Schema, API Documentation, Installation Guide, and a condensed User Manual for the psyclick-secure application.

The full clinician and patient operating guide is provided as a separate User Manual deliverable; the section below summarizes the operating procedure.

2. System Architecture Diagram

PsyClick Secure applies defense in depth within a single-workstation trust boundary. A local Electron/React presentation layer communicates with a loopback Flask API (127.0.0.1:5101); server middleware authenticates bearer sessions; role checks and clinician ownership scope protected records; a fail-closed consent gate precedes telemetry capture; SQLite persistence uses parameterized queries and transactions; audit events are hash-chained; and Windows DPAPI protects exports and backups.

Figure 38. PsyClick Secure — Layered Defense-in-Depth Architecture

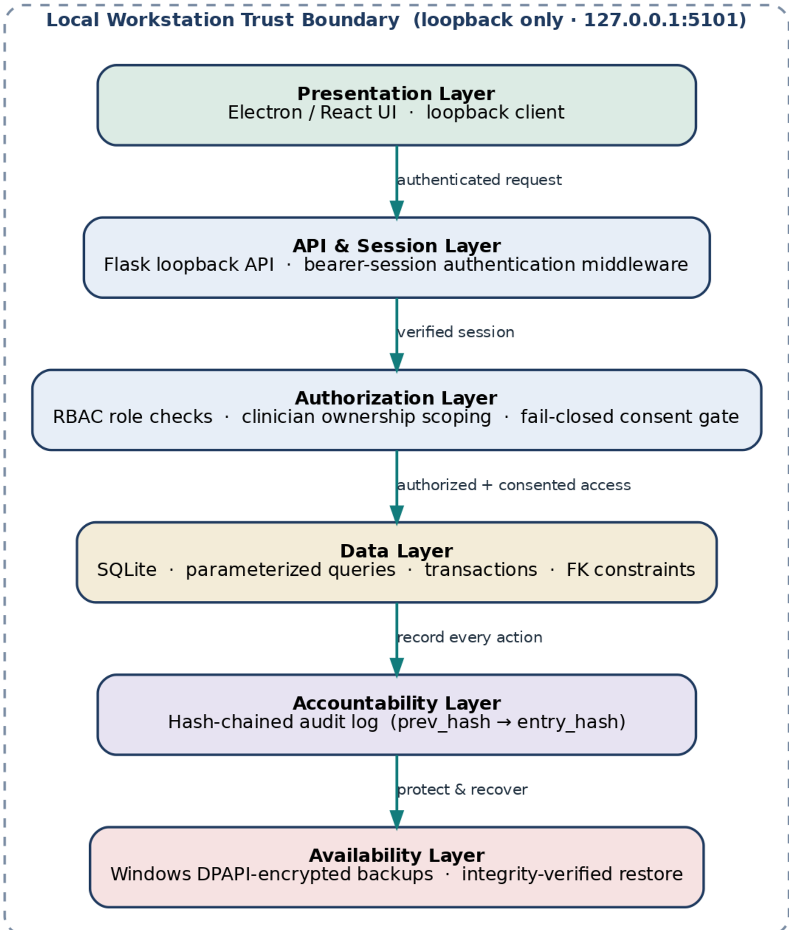

Figure 1. PsyClick Secure layered security architecture.

3. Database Schema

The secure database classifies clinician identity, sessions, consent, clinical/behavioral records, audit evidence, and recovery metadata. Direct SQL string concatenation is prohibited; parameterized queries, foreign-key constraints, transaction boundaries, and application-level authorization are required.

Entity	Primary Key	Security-Relevant Fields	Classification
clinicians	clinician_id	scrypt password_hash; role; status; failed_attempts; locked_until	Restricted identity
security_sessions	session_id	token_hash; clinician_id; role; last_seen_at; expires_at; revoked_at	Confidential session metadata
consent_records	consent_id	patient_id; clinician_id; consent_version; decision; recorded_at; withdrawn_at	Restricted consent

intake_sessions	session_id	patient/session identifiers; features; scores; flag; rationale	Restricted clinical/behavioral
audit_log	log_id	actor; actor_id; action; outcome; prev_hash; entry_hash; timestamp	Confidential accountability
backup_records	backup_id	created_by; file_path; SHA-256; encrypted; status; verified_at	Confidential recovery metadata

The security-aware entity-relationship diagram is shown below.

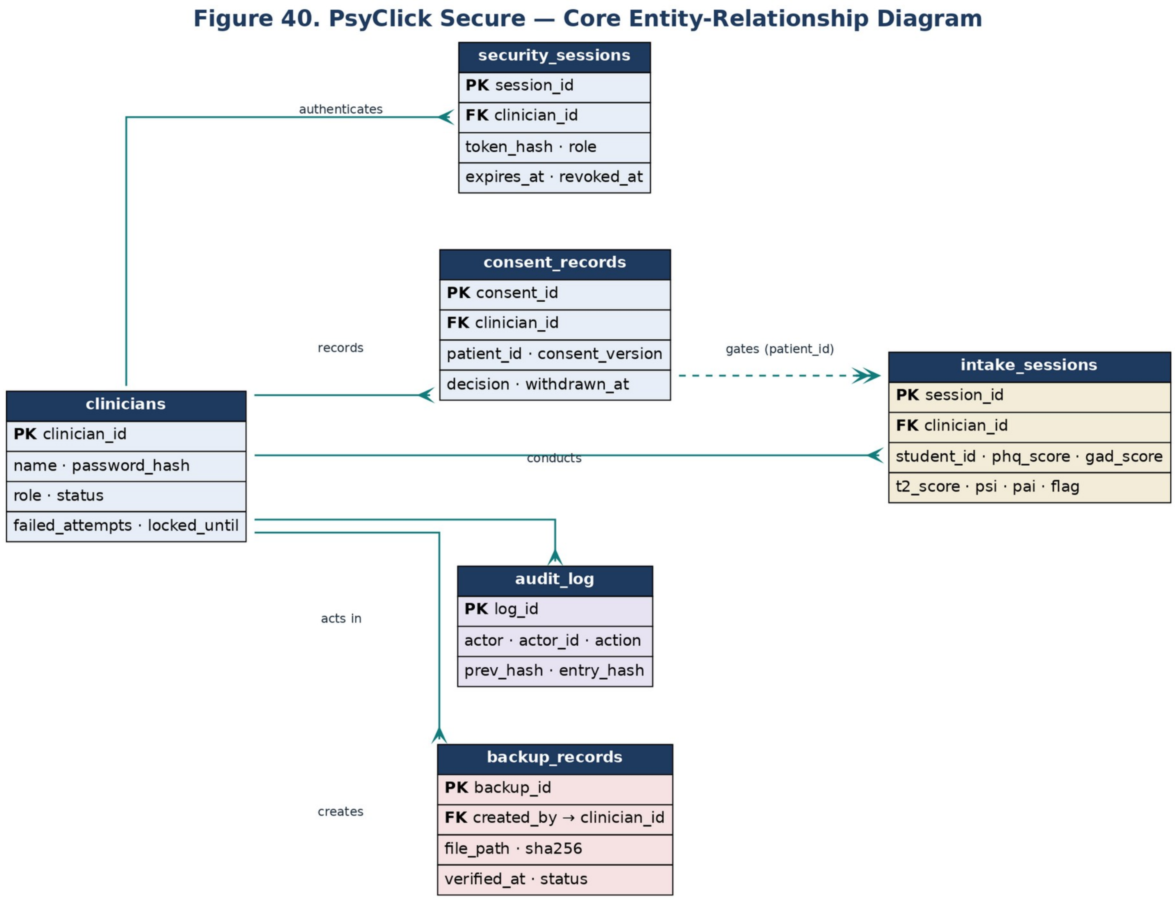

Figure 2. Core entity-relationship diagram.

4. API Documentation

All endpoints are served from the loopback Flask API at http://127.0.0.1:5101/api/ and require an authenticated bearer session except for login, first-account bootstrap registration, and the readiness ping. Access is enforced by role-based middleware.

Authentication

POST /api/login, POST /api/register (bootstrap or admin-invoked), POST /api/verify-clinician, POST /api/logout

Clinical workflow

POST /api/intake/start, GET /api/next-client-id, GET /api/next-tester-id, calibration (keyboard/mouse start and save), PHQ-9/GAD-7 and emotional-task assessment routes, POST /api/assessment/finish

Records and reporting

GET /api/stats, GET /api/sessions/recent, GET /api/session/<id>, GET /api/patients, GET /api/patients/<id>/sessions, POST /api/export/report, POST /api/export/summary

Audit and security monitoring

GET /api/audit (admin, auditor), POST /api/audit/log, GET /api/audit/verify (admin, auditor) — hash-chain verification, GET /api/security/intrusion-alerts (admin, auditor)

Administration

GET /api/admin/users (admin), PATCH /api/admin/users/<id> (admin), POST /api/admin/backup, POST /api/admin/backup/verify, GET /api/admin/security-status (admin)

Sync and health

GET /api/ping, GET /api/db-health, GET /api/sync/status, POST /api/sync/now, POST /api/sync/pull (optional Supabase cloud sync; no-op if unconfigured)

Normative / research mode

POST /api/normative/login, POST /api/normative/mode, GET /api/normative/stats, POST /api/normative/compute, GET /api/normative/compare/<id>

5. Installation Guide

On a supported Windows workstation:

- Create a Python virtual environment and install requirements.txt.
- For a packaged/controlled deployment, set the PSYCLICK_SECURE_HOME environment variable for the secure data directory (defaults to %APPDATA%\PsyClickSecure).
- Build the backend and frontend with BUILD_SECURE.bat (PyInstaller creates dist-python\psyclick_api\psyclick_api.exe; Vite builds the frontend).
- To run in development, start api_server.py on 127.0.0.1:5101, install frontend dependencies, and run npm run dev — or use start_secure.bat.
- To produce the installable desktop app, run npm run dist:installer in the frontend folder.
- Launch the application and create the first administrator account; further accounts require an administrator session.

The separate source archive and frontend build archive are supplied. A fresh independent-workstation installation still requires recorded evidence.

6. User Manual (Summary)

The full Clinician Manual and Patient Guide are provided as a separate User Manual deliverable. The core operating procedure is:

- Sign in with an individual clinician account and confirm the API readiness indicator.
- Create a pseudonymous client and record informed consent (capture is blocked until consent is recorded).
- Run keyboard and mouse calibration, then administer PHQ-9, GAD-7, and the emotional-task prompts.
- Review the dashboard — T^2, PSI/PAI, confidence, and the Green/Amber/Red signal — alongside clinical observation.
- Export only to an approved protected destination; exports and backups are DPAPI-encrypted with a SHA-256 check.
- Review the audit log, log out, and lock the workstation.

7. Approval

Name	Signature

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